Relations between trigonometric ratios

1. $\sin\theta = \frac{1}{\operatorname{cosec}\theta}$ or $\operatorname{cosec}\theta = \frac{1}{\sin\theta}$

2. $\cos\theta = \frac{1}{\sec\theta}$ or $\sec\theta = \frac{1}{\cos\theta}$

3. $\tan\theta = \frac{\sin\theta}{\cos\theta}$ or $\cot\theta = \frac{\cos\theta}{\sin\theta}$

4. $\tan\theta = \frac{1}{\cot\theta}$ or $\cot\theta = \frac{1}{\tan\theta}$

Trigonometric Identities

1. $\sin^2\theta + \cos^2\theta = 1$
2. $1 + \tan^2\theta = \sec^2\theta$
3. $1 + \cot^2\theta = \operatorname{cosec}^2\theta$.

Angle

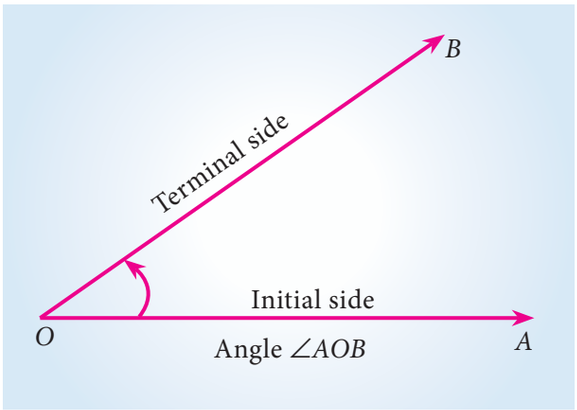


Fig. 4.1

Angle is a measure of rotation of a given ray about its initial point. The ray 'OA' is called the initial side and the final position of the ray 'OB' after rotation is called the terminal side of the angle. The point 'O' of rotation is called the vertex.

If the direction of rotation is anticlockwise, then angle is said to be positive and if the direction of rotation is clockwise, then angle is said to be negative.

Measurement of an angle

Two types of units of measurement of an angle which are most commonly used namely degree measure and radian measure.

Degree measure

If a rotation from the initial position to the terminal position $\left(\frac{1}{360}\right)^{th}$ of the revolution, the angle is said to have a measure of one degree and written as 1°. A degree is divided into minutes and minute is divided into seconds.

One degree = 60 minutes (60′)
One minute = 60 seconds (60″)

Radian measure

The angle subtended at the centre of the circle by an arc equal to the length of the radius of the circle is called a radian, and it is denoted by 1^c

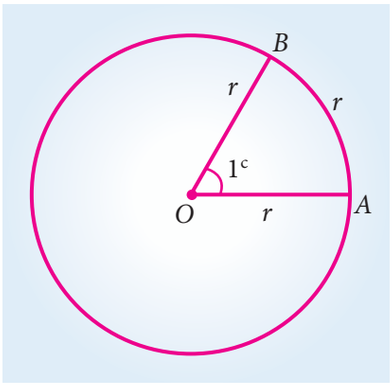


Fig. 4.2

NOTE

The number of radians in an angle subtended by an arc of a circle at the centre of a circle is

$$= \frac{length\ of\ the\ arc}{radius}$$

i.e., $\theta = \frac{s}{r}$, where θ is the angle subtended at the centre of a circle of radius r by an arc of length *s*.

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